PassFrame tuition

Angles & 2D Shapes

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. Reflex

2. Acute

3. C — 135°

4. 180°

5. 360°

6. 7

7. B — nonagon

8. 10

9. A, C and E (200°, 300° and 359° are all between 180° and 360°)

10. 180°

11. Scalene

12. B — regular hexagon

13. Obtuse-angled triangle

14. Reflex angle

15. 3

Standard

1. 96°

2. 140°

3. B — 60°

4. 80°

5. 180° each

6. 84°

7. 63°

8. A, B and E (a parallelogram has equal and parallel opposite sides, and equal opposite angles — it does not need equal sides or right angles)

9. 55° and 70°

10. 62°

11. 105° each

12. A — trapezium

13. 85°

14. 65°

15. 135°

Standard / Hard

1. x=50; angles are 70° and 110°

2. b = 60° (since 2b + b = 180)

3. 6 sides (360 ÷ 60)

4. A, C and E (a kite has two pairs of adjacent equal sides, one pair of equal opposite angles, and at least one line of symmetry)

5. 35° (since 90 − 55)

6. 8 slices (360 ÷ 45)

7. 110°

8. 30°, 60° and 90°

9. 90° each — a square or a rectangle

10. 35° (since 90 − 55)

11. 8 sides (exterior = 45°, 360 ÷ 45)

12. 55°; a right-angled triangle

13. 75°, 105° and 75°

14. 90° (360 − 80 − 80 − 110)

15. 68° each

Homework — Answers

Easy / Standard

1. Reflex

2. Acute

3. D — 220°

6. 7

7. C — decagon

8. 9

11. Isosceles

12. B — regular pentagon

13. Acute-angled triangle

4. 180°

5. 360°

9. A, C and E (190°, 350° and 181° are all between 180° and 360°)

10. 180°

14. Acute angle

15. 4

Standard

1. 84°

2. 140°

3. B — 60°

4. 80°

5. 90° each

6. 56°

7. 51°

8. A, B and D (a rectangle has equal and parallel opposite sides and four right angles — it does not need all sides equal, and a non-square rectangle has only 2 lines of symmetry)

9. 62° and 56°

10. 57°

11. 95° each

12. A — rhombus

13. 100°

14. 58°

15. 108°

Standard / Advanced

1. x=50; angles are 80° and 100°

2. B — 45° (since 3b + b = 180)

3. 8 sides (360 ÷ 45)

4. A, C and E (a trapezium has one pair of parallel sides, is a quadrilateral with 4 sides — it does not need equal sides or right angles)

5. 28° (since 90 − 62)

6. 10 slices (360 ÷ 36)

7. 115°

8. 36°, 54° and 90°

9. A square

10. 42° (since 90 − 48)

11. 10 sides (exterior = 36°, 360 ÷ 36)

12. 48°; a right-angled triangle

13. 62°, 118° and 62°

14. 95° (360 − 70 − 70 − 125)

15. 64° each